

MANIKANTA PUDOKA

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PROFESSIONAL SUMMARY

Results-driven **Data Engineer** with **3+ years of experience** designing, building, and optimizing **data pipelines**, **ETL workflows**, and **analytics ecosystems** across **cloud platforms**. Skilled in **Python**, **SQL**, and **Apache Airflow** for **data ingestion**, **transformation**, and **automation**. Strong background in **data warehousing** (**Snowflake**, **Redshift**, **BigQuery**) and **data modeling** (**Star** and **Snowflake Schema**) to power **BI**, **reporting**, and **machine learning pipelines**. Proficient in developing **interactive dashboards** using **Power BI**, **Tableau**, and **Looker** to translate complex data into actionable insights. Hands-on experience with **AWS**, **Azure**, and **GCP**, ensuring **scalable**, **reliable**, and **secure** cloud data infrastructure. Adept in **data validation**, **pipeline optimization**, and **cross-functional collaboration**, driving **data-driven decision-making** across **engineering**, **research**, and **business teams**.

SKILLS

- **Data Engineering:** SQL (PostgreSQL, Redshift, BigQuery), Python (pandas, pyarrow), ETL & orchestration (Apache Airflow, Azure Data Factory, dbt), Data Warehousing & Modeling (Snowflake, Star/Snowflake Schema), PySpark, Object Storage (AWS S3, GCS), API Integration, Automation Scripts
- **Data Analysis:** Exploratory Data Analysis (EDA), Advanced Excel (Pivot Tables, Formulas), SQL Analytics & Window Functions, Dashboarding & Reporting (Power BI, Tableau, Looker), KPI Design, Trend & Root-Cause Analysis, A/B Testing, Hypothesis Testing, Stakeholder Storytelling
- **Data Science:** pandas, NumPy, scikit-learn, Jupyter, Feature Engineering, Cross-Validation, Regression & Classification Models, Model Evaluation (Precision, Recall, ROC, RMSE), Basic ML Pipelines
- **Cloud & DevOps:** AWS (S3, Lambda), GCP, Azure (Fundamentals), Git, Docker (Basic), Bash, VS Code
- **Databases & Storage:** PostgreSQL, MySQL, Snowflake, Redshift, BigQuery, AWS S3, GCS
- **Visualization & ML Libraries:** Power BI, Tableau, Matplotlib, Seaborn, scikit-learn, MLflow (Basic)

PROFESSIONAL EXPERIENCE

McKinsey & Company, USA | Data Engineer | USA

Sep 2025 – Present

- **Orchestrated** multi-source ETL pipelines using **Apache Airflow**, **enhancing data reliability** and cutting daily job failures from **12 to under 3**, which improved end-to-end **data delivery SLAs** for client analytics projects.
- **Refactored** analytical data models in **Snowflake**, **reducing query latency** from **90s to 28s** and boosting performance for **ad-hoc SQL analytics and executive dashboards** across cross-functional consulting teams.
- **Engineered** reusable transformation scripts in **Python (pandas, pyarrow)** to process **8M+ transactional records monthly**, ensuring **data quality, lineage, and validation** for client engagement databases.
- **Deployed** automated ingestion pipelines into **AWS S3** via API connectors, scaling centralized **data lake storage by 1.5 TB** and improving **data accessibility** for downstream analytics and **machine learning models**.
- **Designed** interactive KPI dashboards in **Tableau**, integrating SQL window functions to **automate performance tracking** for **20+ senior stakeholders**, strengthening **data-driven decision-making** in strategy reviews.

Saint Louis University | Graduate Research Assistant - Health Data Science | USA

Jan 2025 – May 2025

- **Engineered** automated ETL pipelines using **Apache Airflow** to clean, preprocess, and integrate **~1.2M electronic health records (EHRs)**, **shortening monthly refresh cycles by 10 hours** and improving **data reproducibility** across multi-source clinical datasets.
- **Queried** structured hospital data with **SQL (Postgres)** to perform **statistical and trend analysis** on **12,000+ patient encounters**, identifying **factors linked to extended recovery periods** and supporting ongoing **outcome-based healthcare studies**.
- **Developed** predictive classification models using **scikit-learn** with feature engineering and model evaluation (ROC/AUC, precision, recall), **raising readmission prediction accuracy from 0.74 to 0.81**, directly informing **patient risk stratification research**.
- **Designed** interactive **Power BI dashboards** visualizing **population health trends, treatment outcomes, and KPIs**, providing **4 faculty research teams** with real-time access to insights for **evidence-based discussions and publication data preparation**.
- **Structured** and standardized **clinical datasets in Snowflake** using **Star schema data modeling**, **reducing average query time from 18s to 11s** and enhancing **data accessibility** for cross-department collaborations in health data science projects.

Tata Consultancy Services (TCS) | Data Engineer | INDIA

Jul 2021 – Dec 2022

- **Architected** production-grade ETL pipelines in **Apache Airflow**, orchestrating ingestion of **~8M daily records** and improving **data delivery timelines by 15%** for national digital service platforms.
- **Revamped** complex **PostgreSQL queries** and indexing strategies, **reducing query execution time by 35%** while supporting **real-time analytics for financial inclusion dashboards**.
- **Orchestrated** distributed data processing with **PySpark**, automating transformation and quality validation that **eliminated 18+ manual hours per week** and improved **data integrity for operations reporting**.
- **Designed** modular **data marts in dbt**, enforcing **schema consistency, lineage documentation, and test automation** across **five concurrent transformation projects**, enhancing governance standards.
- **Deployed** dynamic **Power BI dashboards**, translating high-volume transactional data into **actionable KPIs** that enabled **regional administrators to cut service turnaround time by 10%**.

Infosys | Data Science, Intern | INDIA

May 2021 – Jul 2021

- **Developed** and fine-tuned regression and classification models using **Python (scikit-learn)**, enhancing data-driven forecasting accuracy by **8%** and enabling better revenue trend predictions for retail analytics teams.
- **Orchestrated** automated **ETL pipelines** through **Apache Airflow**, streamlining data ingestion from Snowflake and APIs, saving **12+ hours weekly** in manual maintenance, and improving overall data reliability.
- **Executed** end-to-end **Exploratory Data Analysis (EDA)** with **pandas and NumPy**, performing data cleaning, preprocessing, and outlier detection that uncovered actionable customer behavior insights.
- **Visualized** operational and financial KPIs using **Power BI**, improving dashboard refresh cycles by **65%**, and helping stakeholders monitor churn, profitability, and performance trends in real time.
- **Assessed** ML model performance in **Jupyter Notebook** using metrics like **ROC-AUC, Precision, and RMSE**, ensuring statistical robustness and model readiness for deployment across multiple datasets.

Adani | Data Analyst | INDIA

May 2020 – May 2021

- **Optimized** large-scale logistics data pipelines using **SQL (PostgreSQL)**, reducing vessel turnaround time by **6 hours** and improving operational efficiency across **4 major ports**.
- **Visualized** renewable energy performance trends in **Power BI**, enhancing KPI transparency and cutting executive reporting cycles by **20%**.
- **Streamlined** cross-system data ingestion with **Apache Airflow**, consolidating **SCADA and ERP feeds** and reducing manual workload by **10–12 hours weekly**.
- **Interpreted** solar plant performance data using **Python (pandas)** and **EDA techniques**, detecting **3 recurring fault patterns** that guided **predictive maintenance planning**.
- **Engineered** and maintained **Snowflake data models (Star Schema)** for financial analytics, **boosting query speed by 35%** and accelerating revenue forecasting for the energy division.

EDUCATION

Saint Louis University | USA

Master of Science in Health Data Science

Jan 2023 – May 2025